Strongly quasisymmetric vacuum fields on a given solid torus

Corrected research brief for Gemini's research agents
Revised: 30 October 2026.
The embedded Euclidean solid torus is input data. Its associated product complex surface is fixed.
Status: exact reformulations, proved elementary obstructions, and research assignments. No solution of the full stellarator target is claimed.

0. Correction and the actual question

The earlier version incorrectly made the embedding of the solid torus an unknown and proposed varying its shape and auxiliary complex geometry as construction variables. That changed the problem. This version supersedes those proposals.

Let a smooth, compact, connected solid torus

$$\Omega \subset \mathbb{R}^3$$

be **given with its embedding and induced Euclidean metric** g. Retain the magnetic-boundary requirement $B \cdot n = 0$ on $\partial\Omega$. Fix a longitudinal generator γ of $H_1(\Omega;\mathbb{Z})$ and a nonzero circulation P.

The first decisive observation is:

> There is exactly one curl-free, divergence-free field tangent to the given boundary with circulation P. The remaining problem is whether this particular field has the required nonvanishing, nested flux surfaces, strong quasisymmetry, and rotational transform.

Thus the holomorphic formulation cannot select a different physical field on the same domain with the same boundary condition and normalization. It can provide a useful representation, an obstruction, or a method to recover additional structures of that field.

The concrete research question is:

> For the normalized harmonic field of the given Euclidean solid torus, find a necessary-and-sufficient criterion for the full strong-QS package below, and determine whether holomorphic or dimensional-reduction methods make that criterion tractable.

No numerical boundary description is supplied here. The brief treats Ω as a fixed input, rather than deciding the answer for an unspecified particular shape.

Fixed data and unknowns

Object	Status
$\Omega \hookrightarrow \mathbb{R}^3$, its boundary and Euclidean metric g	Given; cannot vary
Circulation $P \neq 0$, homology marking, orientation	Given; choose one field normalization
$X = \Omega \times \mathbb{R}_t$, $g_0 = g + dt^2$, standard I_0, ω_0 below	Fixed by the specified product construction
Vacuum field $B = B_P$	Uniquely determined by the first two rows
Curvature representing B	Uniquely determined; flat connection additions do not change it
Strong-QS generator u, moment function Ψ, axis, internal flux surfaces	To be found, or proved not to exist
Auxiliary quotients, holomorphic descriptions, higher-rank data	Permitted as methods, with reconstruction to the same fixed data

The magnetic boundary is an outer flux surface of the confined core. If instead Ω were merely a vessel and the normal field on its boundary were unspecified, the uniqueness statement would not apply. That is a different boundary-value problem; it is not adopted here.

1. The normalized vacuum field is already determined

Write

$$\nu = \mathrm{vol}_g, \qquad \alpha = B^\flat, \qquad \beta = \iota_B \nu = *_3 \alpha.$$

The physical vacuum problem is

$$\boxed{\operatorname{curl} B = 0, \quad \operatorname{div} B = 0, \quad B \cdot n = 0 \text{ on } \partial\Omega, \quad \oint_\gamma \alpha = P.} \tag{V}$$

All operators in this document use the given Euclidean metric unless another metric is explicitly labeled auxiliary.

Proposition 1.1: existence and uniqueness

Choose a smooth closed one-form η with $\oint_\gamma \eta = 1$, and put $v_0 = \eta^\sharp$. Solve the scalar Neumann problem

$$\Delta f = -P \operatorname{div} v_0, \qquad \partial_n f = -P v_0 \cdot n, \qquad \int_\Omega f \, \nu = 0. \tag{N}$$

The compatibility condition follows from the divergence theorem. Then

$$\alpha_P = P\eta + df, \qquad B_P = \alpha_P^\sharp$$

solves (V). Its smoothness follows from the smooth Neumann problem.

For uniqueness, the difference of two solutions has a closed dual one-form with zero period on the generator of $H_1(\Omega)$, hence equals dq. The vacuum and boundary equations give $\Delta q = 0$, $\partial_n q = 0$. Integration by parts gives $dq = 0$.

Equivalently, the space of tangent harmonic fields is one-dimensional. This standard fact also underlies [R7]. The displayed argument is included to make its boundary hypotheses explicit.

A nonzero period ensures that B_P is not identically zero. It does **not** by itself establish pointwise nonvanishing, an invariant torus foliation, or quasisymmetry. Those are separate tests.

Fixing a nonzero meridional flux instead of circulation is an equivalent normalization. Their ratio is determined by the fixed domain; they are not independent adjustable parameters. Scaling the field cannot repair a failure of the qualitative QS or foliation requirements.

A given domain need not pass the target

For the round torus

$$\Omega = \{(r, \phi, z) : (r - R)^2 + z^2 \le a^2\}, \qquad R > a,$$

the unique field is

$$B_P = \frac{P}{2\pi r} \, e_\phi, \qquad \alpha_P = \frac{P}{2\pi} \, d\phi.$$

It is nonvanishing and vacuum, but its field lines have zero poloidal winding in the usual meridional/preferred-longitudinal marking. It is also spatially axisymmetric. It therefore fails the full nonaxisymmetric, nonzero-transform target. This is a counterexample to a universal positive assertion for every given solid torus, not a nonexistence theorem for all shapes.

2. The remaining target: global strong quasisymmetry

Here "strong" has its conventional meaning for the usual truncated guiding-centre model [R1–R3]. It is not an assertion of exact symmetry of full Lorentz-force dynamics at every gyroradius. Requiring a Euclidean Killing symmetry would impose an additional condition and exclude the intended hidden symmetries.

First require $B = B_P$ to be nonvanishing on the whole compact domain. Let $Q = |B|^2 > 0$.

Seek a smooth vector field u and real function Ψ such that

$$\boxed{\iota_u \beta = d\Psi, \qquad \mathcal{L}_u \alpha = 0, \qquad \mathcal{L}_u \nu = 0.} \tag{SQS}$$

These equations are supplemented by the following global requirements.

1. **Nested surfaces and axis.** After choosing the sign and additive constant of Ψ, its minimum set is a smooth circle Γ, its regular levels are connected embedded tori, and $\partial\Omega = \Psi^{-1}(\Psi_b)$. Require a Morse–Bott minimum along Γ, with positive normal Hessian, and no other critical points. The axis is a closed field line. All physical fields extend smoothly across it.
2. **A periodic symmetry.** The flow of u defines an effective circle action, normalized to period 2π on generic orbits. Fixed or exceptional orbits must be treated explicitly, including smooth axis extension. On regular flux surfaces, B and u are independent.
3. **Nonzero rotational transform.** In compatible straight-field-line angles, $B\theta/B\phi = \iota(\Psi)$ with $B\phi \neq 0$, and $\iota \neq 0$ throughout the regular core. The meridional and longitudinal marking is fixed by the embedded configuration; changing the longitude to shift the winding ratio by an integer does not create the requested transform. A nonconstant continuous transform cannot be irrational on every surface.
4. **Genuine lack of spatial symmetry.** No nontrivial one-parameter Euclidean isometry group preserves the domain and its field. Exhibiting one non-Killing u does not establish this: a symmetric configuration can admit additional non-Killing generators.

Because (V) fixes the normalized field, every continuous Euclidean symmetry of the domain preserves that field. The symmetry of the given domain is therefore already a useful preliminary test of requirement 4.

These requirements concern the vacuum core. They do not include finite-coil realization, finite-pressure equilibrium, stability, turbulent transport, or exhaust.

Immediate identities

Cartan's formula gives $\mathcal{L}_u \beta = 0$. Together with volume preservation this implies

$$[u, B] = 0, \qquad B\Psi = u\Psi = 0, \qquad uQ = 0.$$

Since $d\alpha = 0$, the one-form equation in (SQS) is precisely

$$\boxed{B \cdot u = C \quad \text{for a single constant } C.} \tag{C}$$

Do not replace this by a freely chosen flux function. The branch $C = 0$ is not excluded by definition.

Circle periods constrain the constant

If a generic u-orbit, parametrized for time 2π, represents $m[\gamma]$ in $H_1(\Omega; \mathbb{Z})$, then

$$2\pi C = \oint_{\text{orbit}} \alpha = mP, \qquad C = \frac{mP}{2\pi}, \quad m \in \mathbb{Z}. \tag{Per}$$

This is an orbit-homology condition, not magnetic flux quantization. The winding within each flux torus carries additional information. Flux-dependent rescaling of u is not harmless: it can destroy the constant in (C) and the common action period.

Boozer-coordinate normalization check

On a regular region admitting suitable Boozer coordinates, use

$$\beta = d\psi \wedge (d\theta - \iota(\psi)d\phi), \qquad \alpha = I(\psi)d\theta + G(\psi)d\phi + K\,d\psi.$$

For $|B| = \mathcal{B}(\psi, M\theta - N\phi)$ and the periodic candidate $u = N\partial_\theta + M\partial_\phi$,

$$\iota_u\beta = (M\iota - N)d\psi = d\Psi, \qquad \Psi(\psi) = \int^{\psi} (M\iota(s) - N)\,ds.$$

Thus the moment Ψ is generally not the normalized toroidal flux ψ. Vacuum makes I, G constant; smooth extension through a solid-torus axis gives $I = 0$. Also $C = NI + MG$. The regular target excludes $M\iota - N = 0$. These are consistency checks when the coordinates exist, not a claim of an unconditional equivalence in every degenerate case.

3. A direct reduction for the fixed field

This reduction should be attempted before introducing another parent gauge theory. Its coefficients are determined by the given domain through (N).

Proposition 3.1: a linear first-order system for the moment

For a known nonvanishing vacuum field B, (SQS) is equivalent to finding (Ψ, C), with C constant, satisfying

$$\boxed{\begin{aligned} B \cdot \nabla\Psi &= 0, \\ (B \times \nabla Q) \cdot \nabla\Psi &= -C\,B \cdot \nabla Q, \qquad Q = |B|^2, \end{aligned}} \tag{FQS}$$

and reconstructing

$$\boxed{u = \frac{\nabla\Psi \times B + CB}{Q}.} \tag{U}$$

Proof. The contraction equation is $B \times u = \nabla\Psi$. Together with $B \cdot u = C$, it gives (U) and $B\Psi = 0$. Conversely these identities follow from (U) when $B\Psi = 0$. Using $\operatorname{curl} B = \operatorname{div} B = 0$,

$$\operatorname{div} u = -\frac{(B \times \nabla Q) \cdot \nabla\Psi + C\,B \cdot \nabla Q}{Q^2}.$$

Thus the second equation is precisely volume preservation. Finally $\mathcal{L}_u\alpha = d(B \cdot u) = dC = 0$. This proves the equivalence of the differential equations.

For fixed B, (FQS) is linear in (Ψ, C). The global conditions in Section 2 remain substantial: regular torus levels, a smooth axis, periodicity of (U), marking, and nonzero transform are not automatic consequences of linearity.

Use $\Psi|_{\partial\Omega} = \Psi_b$. With the magnetic boundary condition this also makes reconstructed u tangent to the boundary. The formula divides only by Q. Dividing instead by $B \cdot \nabla Q$ or $|\nabla Q|$ can discard valid degenerate branches.

A local obstruction involving finitely many derivatives

Set

$$X_B = B, \qquad Y_B = B \times \nabla Q, \qquad h = BQ, \qquad Z_B = [X_B, Y_B].$$

Equation (FQS) implies

$$X_B\Psi = 0, \qquad Y_B\Psi = -Ch, \qquad Z_B\Psi = -CX_Bh.$$

On an open set where X_B, Y_B, Z_B are independent, define the uniquely determined one-form η_B by

$$\eta_B(X_B) = 0, \qquad \eta_B(Y_B) = -h, \qquad \eta_B(Z_B) = -X_Bh. \tag{Jet}$$

Then:

- If $C = 0$, independence forces $d\Psi = 0$. Such a region cannot belong to the regular flux-surface part of the target.
- If $C \neq 0$, necessarily $d\Psi = C\eta_B$, hence $d\eta_B = 0$.
- Conversely, on a simply connected patch in this rank-three region, $d\eta_B = 0$ allows integration of $d\Psi = C\eta_B$ and gives a local solution of (SQS) through (U). Nondegeneracy requires $\eta_B \neq 0$. No global periodicity or torus topology follows from this local result.

Thus a rank-three regular point with $d\eta_B \neq 0$ is a concrete local obstruction. Rank-deficient loci require a separate analysis; this test must not be used as a global criterion without that work.

An exact local check. The vacuum field $B = \nabla(xyz) = (yz, xz, xy)$ has, at $(1, 2, 3)$,

$$Q = 49, \quad X_B = (6, 3, 2), \quad Y_B = (10, -128, 162), \quad Z_B = (240, -768, 432).$$

Their determinant is 564480. At that point,

$$\eta_B = -\frac{2201}{720}dx + \frac{869}{180}dy + \frac{463}{240}dz,$$

and the $dx \wedge dy$ coefficient of $d\eta_B$ is $1739/21600 \neq 0$. These values follow by differentiating the defining linear system (Jet), and have been checked using exact rational arithmetic. This is a local vacuum example demonstrating that the test is not an identity satisfied by every harmonic field. It is not a tangent-boundary solid-torus counterexample.

The calculation is supplied for independent checking, without a novelty claim.

An alternative centralizer formulation

One can instead solve for u using

$$[u, B] = 0, \qquad \operatorname{div} u = 0, \qquad d(B \cdot u) = 0, \qquad [\iota_u\beta] = 0 \in H^1(\Omega; \mathbb{R}). \tag{Cent}$$

The first two conditions make $\iota_u\beta$ closed. Its remaining cohomology class is tested by its longitudinal period. Exactness then supplies a global Ψ. Boundary tangency and all global conditions still have to be imposed. Finding a commuting vector field alone is not sufficient.

4. The line bundle and the fixed complex surface

Three-dimensional connection

Since $H^2(\Omega; \mathbb{R}) = 0$, choose a global real one-form a with $da = \beta$. On the trivial smooth Hermitian line bundle use

$$\nabla = d + ia, \qquad F_\nabla = i\beta.$$

Then (V), apart from its circulation normalization, is

$$d\beta = 0, \qquad d(*_3\beta) = 0, \qquad j^*\beta = 0, \quad j : \partial\Omega \hookrightarrow \Omega.$$

The boundary pullback is exactly the magnetic boundary condition. Flat additions to a can change connection holonomy while leaving B unchanged. Ordinary absolute Chern-class integrality places no restriction on the continuous circulation P here.

Strong QS gives an infinitesimal connection symmetry,

$$\mathcal{L}_u a = d(\Psi + a(u)).$$

On the principal circle bundle with connection Θ, the lifted field $u^H - \Psi\partial_\vartheta$ preserves Θ. A globally periodic lift requires additional orbit/holonomy checks.

The prescribed product geometry

With the given Cartesian coordinates (x_1, x_2, x_3), set

$$X = \Omega \times \mathbb{R}_t, \qquad z_1 = x_1 + ix_2, \quad z_2 = x_3 + it,$$

$$g_0 = g + dt^2, \qquad \omega_0 = dx_1 \wedge dx_2 + dx_3 \wedge dt, \qquad \nu_4 = \nu \wedge dt.$$

Use the standard complex structure I_0. Once this product construction has been specified, X, I_0, g_0, ω_0 are all fixed. Choosing a different complex surface or Kähler metric as the physical base is outside the problem.

Define

$$\mathcal{F} = \beta + dt \wedge \alpha, \qquad \widehat{\nabla} = d + i(a + t\alpha), \qquad F_{\widehat{\nabla}} = i\mathcal{F}. \tag{Lift}$$

For the displayed orientation,

$$*_4\mathcal{F} = -\mathcal{F}, \qquad d\mathcal{F} = 0 \Longleftrightarrow d\beta = d\alpha = 0.$$

On this Kähler surface, a real anti-self-dual two-form is primitive and of type $(1,1)$. Therefore the vacuum field gives

$$\boxed{F^{0,2}_{\widehat{\nabla}} = 0, \qquad i\Lambda_{\omega_0} F_{\widehat{\nabla}} = 0.} \tag{HYM}$$

The first equation defines a holomorphic structure on the smooth line bundle; the second is the Hermitian–Yang–Mills equation with zero Einstein constant. No compact-manifold slope theorem is being applied. Smooth topological triviality does not authorize assuming a global holomorphic frame. No Stein or pseudoconvex-boundary hypothesis is being inserted.

Locally, in a holomorphic frame with Hermitian metric h, the Chern curvature is $-\partial\bar{\partial}\log h$. The primitive-curvature equation is a scalar harmonic equation for $\log h$ in that frame. This local simplification does not encode the boundary, period, or strong-QS constraints by itself.

Converse and uniqueness at the curvature level

Let $T = \partial_t$. For a translation-invariant closed ASD real two-form on this fixed product, put

$$\alpha = \iota_T \mathcal{F}, \qquad \beta = \mathcal{F} - dt \wedge \alpha.$$

These are spatial forms, and ASD gives $\beta = *_3 \alpha$. Closure gives the two vacuum equations. Adding the spatial magnetic boundary condition and period P therefore reconstructs exactly B_P.

Consequently, all admissible line-bundle solutions of (HYM) with these additional conditions have the same physical curvature. Varying the holomorphic description does not evade Proposition 1.1.

The full fixed-surface formulation

Seek a unitary connection on the smooth trivial line bundle over the fixed X, a real vector field U, a real function Ψ, and a constant C, satisfying

$$\boxed{\begin{array}{llll} F_{\widehat{\nabla}} = i\mathcal{F}, & F^{0,2}_{\widehat{\nabla}} = 0, & i\Lambda_{\omega_0} F_{\widehat{\nabla}} = 0, & \\ \mathcal{L}_T \mathcal{F} = 0, & [T, U] = 0, & dt(U) = 0, & T\Psi = 0, \\ \iota_U \mathcal{F} = d(\Psi - Ct), & \mathcal{L}_U \nu_4 = 0. & & \end{array}} \tag{HQS}$$

Impose, in addition:

- $\alpha = \iota_T \mathcal{F}$, $\oint_\gamma \alpha = P$, and $\alpha \neq 0$ everywhere;
- $j^*\beta = 0$, where $\beta = \mathcal{F} - dt \wedge \alpha$ and j is the spatial boundary inclusion;
- all the global conditions in Section 2 for $B = \alpha^\sharp$, $u = U|_{t=0}$, and Ψ.

Indeed,

$$\iota_U \mathcal{F} = \iota_u \beta - \alpha(u) dt,$$

so the last line of (HQS) is exactly the contraction equation, the constant-dot-product condition, and volume preservation. This establishes equivalence with the stated physical target, rather than merely an implication.

Do **not** impose vanishing of the pullback of the entire $\mathcal{F}$ to ∂X: the mixed $dt \wedge \alpha$ term need not vanish there.

For a nonzero field,

$$\mathcal{F} \wedge \mathcal{F} = -2Q\, \nu_4.$$

Thus $\mathcal{F}$ is itself symplectic, with the opposite orientation to ω_0. Equation (HQS) makes U Hamiltonian for $\mathcal{F}$, with moment $\Psi - Ct$. It does not make U Hamiltonian for ω_0, or holomorphic for I_0. The distinction is essential.

Translation and global monodromy

A globally exact $\alpha = d\Phi$ would make Φ a harmonic Neumann function and force $B = 0$. Instead, on the spatial universal cover,

$$\alpha = d\Phi, \qquad \Delta\Phi = 0, \qquad \Phi \circ \gamma = \Phi + P. \tag{Aff}$$

Locally, subtracting $d(t\Phi)$ from the potential $a + t\alpha$ gives $a - \Phi dt$, the usual abelian Bogomolny form. Globally, Φ is affine under the deck transformation, not a single-valued real Higgs field on Ω.

The curvature is translation-invariant, but translation by s changes $a + t\alpha$ by $s\alpha$. This is generally not a global gauge transformation: its longitudinal period is sP. Thus invariance of curvature must not be confused with an invariant connection modulo gauge for every real s.

If one introduces an auxiliary compactification $t \sim t + T_0$, unitary gluing requires $T_0 P/(2\pi) \in \mathbb{Z}$, in the chosen curvature units; the resulting line bundle may be topologically nontrivial. The specified formulation uses $\mathbb{R}_t$, so this compactification is not part of the input. Likewise, one cannot discard the physical boundary by applying a closed-manifold theorem to a different base.

5. Why the straightforward KYM-to-GV analogy fails

What the established reduction actually does

The standard gravitating-vortex construction starts with a different geometric setup: a compact Riemann surface times $\mathbb{P}^1$, an $SU(2)$-equivariant rank-two holomorphic bundle determined by a line bundle and section on the curve, and a coupled metric/connection problem [R6a]. Equivariance reduces KYM to vortex and scalar-curvature equations on the curve.

Writing that curve as Σ, its line metric as h, and its holomorphic section as φ, the reduced equations are

$$\bar{\partial}_L \varphi = 0, \qquad i\Lambda_{\omega_\Sigma} F_h + \tfrac{1}{2}(|\varphi|_h^2 - \tau) = 0,$$

$$S_{\omega_\Sigma} + \kappa(\Delta_\Sigma + \tau)(|\varphi|_h^2 - \tau) = c. \tag{GV}$$

Here the scalar-curvature and Laplacian conventions are those of [R6a–R6b]. Both the curve's metric and the line metric participate in the equations. The holomorphic section is additional Higgs data; it is not the affine magnetic potential Φ in (Aff).

That theorem does not apply directly to our fixed product domain. Its $\mathbb{P}^1$ factor, equivariance, Higgs section, and variable metric are additional structures. None supplies the required magnetic boundary, affine vacuum period, or hidden symmetry of the given B_P.

Proposition 5.1: obstruction for the direct flat rank-one KYM ansatz

Using the convention in [R6b], the standard coupled equations have the form

$$i\Lambda F = \lambda, \qquad S_{g_0} - \kappa\Lambda^2(F \wedge F) = c, \tag{KYM}$$

where $\kappa \neq 0$ and c are constants. For our rank-one lift, $\lambda = 0$, $S_{g_0} = 0$, and

$$F \wedge F = 2Q\nu_4, \qquad \Lambda^2(F \wedge F) = 4Q.$$

The scalar equation therefore forces Q constant. Vacuum in Euclidean space gives $\Delta B = 0$, so

$$\Delta|B|^2 = 2|\nabla B|^2.$$

Consequently B would be a constant vector. A nonzero constant vector cannot be tangent to the entire boundary of a bounded domain, because

$$|B|^2 \operatorname{Vol}(\Omega) = \int_\Omega \operatorname{div}((B \cdot x)B)\, dV = \int_{\partial\Omega} (B \cdot x)(B \cdot n)\, dS = 0.$$

This rules out the **direct, flat, rank-one, nonzero-coupling KYM ansatz**, even before asking for QS. The calculation remains valid with the domain fixed. It does not rule out every higher-rank auxiliary parent system.

Setting $\kappa = 0$ discards this scalar coupling. Replacing the constant right-hand side by a tailored function changes the system and needs an independently justified purpose.

Proposition 5.2: a horizontal holomorphic QS generator is too restrictive

Suppose, in addition to (HQS), that $U = (u_1, u_2, u_3, 0)$, independent of t, is real holomorphic for the **given** I_0.

Its complex components are $f_1 = u_1 + iu_2$ and $f_2 = u_3$. Since f_2 is real-valued and holomorphic, it is constant. Holomorphicity in z_2, together with t-independence, makes f_1 independent of x_3. Volume preservation gives

$$2\,\mathrm{Re}\, f_1'(z_1) = 0.$$

Thus, locally and hence on the connected domain,

$$f_1(z_1) = iaz_1 + b, \qquad a \in \mathbb{R},\ b \in \mathbb{C}, \qquad u_3 = c_3 \in \mathbb{R}.$$

This is a Euclidean Killing field. A global tangent generator satisfying (SQS) also preserves B, and therefore contradicts the required absence of continuous spatial symmetry.

The obstruction concerns this additional holomorphic-generator ansatz. The connection may be holomorphic while its QS generator is not. That is the formulation retained in (HQS).

Adding a vertical component is not a free repair. If $W = U + fT$, then for a QS solution with translation-invariant curvature,

$$\mathcal{L}_W \mathcal{F} = df \wedge \alpha.$$

A curvature-preserving holomorphic lift of this kind must satisfy that extra equation as well as its holomorphicity and global reconstruction conditions.

Changing the metric is not an allowed physical escape

ASD is conformally invariant in four dimensions, but the physical metric and complex surface here are fixed. Any different metric can only be an auxiliary computational device and must reconstruct the same normalized curvature of Section 1.

Even the simplest auxiliary proposal is restrictive: if $\hat{g} = e^{2f} g_0$ is Kähler for the same I_0, then

$$0 = d(e^{2f}\omega_0) = 2e^{2f} df \wedge \omega_0$$

forces $df = 0$. Allowing another complex structure introduces a different auxiliary problem; it supplies no permission to change the prescribed X, I_0 in (HQS), and cannot change the unique physical B_P.

6. Reductions that remain relevant on the fixed domain

The purpose of another reduction must now be to decide compatibility or uncover structure of B_P. It cannot be to tune the physical field through new geometric moduli while retaining (V).

A. The fixed-field PDE and its local obstruction

Use (N) to obtain B_P, then study (FQS), the rank-three test (Jet), and the centralizer formulation (Cent).

Concrete objectives: characterize rank-deficient branches; find coordinate-invariant prolongation conditions; determine how local moments glue across these branches; impose their periods and the circle-action condition. If using numerics, aim for certified nonvanishing and certified obstruction bounds, rather than treating a small QS residual as an exact solution.

This is the most direct approach. It already isolates the new unknowns without changing any physical geometry.

B. A genuine reduction to a disk return map

If B_P is nonvanishing, choose the orientation of γ so $P > 0$. The closed form α/P has integral periods and defines

$$p : \Omega \longrightarrow \mathbb{R}/\mathbb{Z}, \qquad dp = \alpha/P$$

in local circle coordinates. Both p and its boundary restriction are submersions: $\alpha \neq 0$, and on the boundary $\alpha(B) = Q > 0$ with B tangent.

The compact submersion theorem gives a fibration. The induced map on fundamental groups is an isomorphism, so its fiber is a connected simply connected compact surface with boundary, hence a disk. Moreover,

$$Bp = Q/P > 0.$$

Thus the fixed field has a global transverse disk and a well-defined first-return map $\mathcal{P}$. This map preserves the nondegenerate area form obtained by restricting β to that disk, since $\mathcal{L}_B\beta = 0$.

A global Ψ restricts to a first integral of $\mathcal{P}$. The desired nested tori cut the disk in nested closed curves around the axis point. This is an exact reduction of part of the global dynamics, with the given boundary and period retained.

Concrete objectives: express the remaining equations of (FQS), circle-action periods, and the marked transform in return-map and return-time data. An integrable area-preserving map is not, by itself, a proof of strong QS. A general area-preserving map is also not holomorphic in the induced conformal structure. Any complex-analytic encoding must account for that distinction explicitly.

C. Reduction by the sought circle action, without making it an isometry

On a region with a free periodic action, let $\pi : \Omega_{\text{reg}} \to S$ be its orbit surface and choose an invariant circle connection ϑ with $\vartheta(u) = 1$. Write

$$d\vartheta = \pi^*\kappa, \qquad \alpha = C\vartheta + \pi^*\lambda, \qquad \beta = \vartheta \wedge \pi^* d\bar{\Psi} + \pi^*\sigma. \tag{Quot}$$

Here λ is a one-form and σ, κ are two-forms on the two-dimensional quotient. Invariance and $\iota_u\beta = d\Psi$ justify these decompositions. Vacuum closure gives

$$d\lambda = -C\kappa.$$

Closure of the displayed β is automatic on a two-dimensional base. Volume preservation means $\iota_u\nu = \pi^*\mu$ for a quotient area form, with $\nu = \vartheta \wedge \pi^*\mu$.

The essential remaining equation is the **pointwise equation with the fixed physical Hodge star**:

$$*_g(C\vartheta + \pi^*\lambda) = \vartheta \wedge \pi^* d\bar{\Psi} + \pi^*\sigma. \tag{Metric}$$

Because the action need not preserve g, this equation can retain orbit-angle dependence. Averaging it loses physical constraints. If $C \neq 0$, one can take $\vartheta = \alpha/C$, a flat invariant connection, which simplifies the formulas but does not remove (Metric).

Concrete objective: determine when (Metric) closes as a useful system on the quotient while retaining every angle-dependent residual. This is a real geometric reduction analogous in organization to an equivariant gauge reduction. It is not automatically a holomorphic quotient of the fixed X. Exceptional or fixed orbits and axis extension require separate treatment.

D. Abelian Bogomolny/minitwistor methods with affine monodromy

Use (Aff) and

$$\beta = *d\Phi, \qquad \Delta\Phi = 0$$

on the spatial cover. These are the abelian Bogomolny equations, already obtained from the fixed four-dimensional ASD connection. Euclidean minitwistor methods relate suitable local monopole data to holomorphic data on the space of oriented lines, modeled by $T\mathbb{P}^1 \simeq \mathcal{O}(2)$ [R9].

Concrete objectives: encode the additive period P, the boundary condition of the given torus, and the reconstruction of its unique harmonic field; then translate (FQS) or (Jet) into a condition on that holomorphic data.

A nonconvex domain may meet a line in several intervals, so a global twistor correspondence for this boundary problem needs proof. The hidden symmetry need not act holomorphically on minitwistor space. A spectral curve alone does not establish the physical boundary, nonvanishing, axis, or transform. This direction is useful only if it provides additional information about the same B_P, beyond a new representation of Maxwell's equation.

E. Elliptic curves supplied by actual internal flux tori

If the required foliation exists, every oriented flux torus inherits a Riemann-surface structure from its induced metric. Since β restricts to zero on a magnetic surface, ∇ restricts to a unitary flat connection, hence a degree-zero holomorphic line bundle.

This gives a family of elliptic curves with points in their Pic^0. On a regular shell, a potential of the form

$$a = \psi\, d\theta - \chi(\psi) d\phi, \qquad \chi' = \iota$$

illustrates how marked angular holonomies encode flux and transform.

Concrete objective: express the fixed field's transverse compatibility and strong-QS equations in this family. The curves, metrics, and their position inside Ω must come from the actual foliation; they are not independent moduli to prescribe. Uniformizing individual tori does not prove the foliation exists, and these tori are not being asserted to be holomorphic curves in the fixed product X.

F. Higher-rank or Higgs parent systems: a conditional research direction

For a higher-rank ASD unitary connection, the determinant curvature is abelian ASD. If proposed as the physical curvature, it must equal the uniquely determined curvature of B_P after normalization. Traces of commutators vanish, so a nonabelian reduction does not automatically enforce QS in the determinant sector. Projection to an eigenline can introduce second-fundamental-form terms that spoil the vacuum equation.

Concrete objective: propose a parent system on a specified auxiliary space and prove that its solvability is equivalent to a missing compatibility condition on this fixed B_P, or gives a new sufficient criterion for that condition. Identify exactly how its additional variables enforce (FQS), periodicity, or the global foliation. A standard KYM, Hitchin, or vortex existence theorem without this reconstruction is insufficient.

This is more speculative than A–C. Varying the physical boundary, Euclidean metric, or prescribed complex surface is not an admissible success in this assignment.

7. Retained counterexample: why abstract flatness is insufficient

The earlier abstract-flat example is still a useful check, but the fixed-domain problem already excludes it at the outset. It is not a reason to introduce a new embedding problem.

Consider

$$M_{\gamma_0,L} = (D^2 \times \mathbb{R})/((z,s) \sim (e^{i\gamma_0} z, s + L))$$

with its flat product metric, $L > 0$, and $\gamma_0/(2\pi)$ irrational. The parallel field $B = \partial_s$ is nonvanishing, vacuum, and tangent to the boundary. Radius levels are nested tori. In periodic angles

$$\theta = \arg z - \gamma_0 s/L, \qquad \phi = 2\pi s/L,$$

its transform is $-\gamma_0/(2\pi)$. The generator

$$u = \frac{\gamma_0}{2\pi}\partial_{\arg z} + \frac{L}{2\pi}\partial_s$$

has period 2π, satisfies $B \cdot u = L/(2\pi)$, and has

$$\iota_u \beta = -\frac{\gamma_0}{2\pi} r\, dr.$$

It supplies strong QS in this abstract geometry. A same-dimensional isometric embedding into Euclidean three-space would turn its parallel field into a nonzero constant vector tangent to a bounded boundary, contradicting Proposition 5.1's elementary boundary argument. It is therefore not the given physical domain. It also has continuous spatial symmetries in its own geometry.

Keep the example as a warning against substituting an abstract flat quotient for the prescribed Euclidean solid torus, not as a candidate solution of the present task.

8. Assignments for the receiving Gemini team

These are proposed tasks for the receiving team, not claims that these investigations have been completed. Each agent must keep $\Omega, g, X, I_0, \omega_0$ fixed.

Assignment	Required output
1. Equivalence and topology audit	`01_audit.md`: check the Neumann uniqueness argument, all Hodge signs, (HQS), monodromy, magnetic boundary, and the precise strong-QS definition. Identify every additional global hypothesis.
2. Fixed-field obstruction analysis	`02_fixed_field_pde.md`: audit (FQS) and (Jet); analyze rank-deficient sets and gluing; establish a useful obstruction or local sufficiency theorem with precise hypotheses.
3. Global dynamics and quotient reduction	`03_global_reduction.md`: develop the disk return-map and/or circle-quotient formulation; retain return times, pointwise metric residuals, orbit homology, axis regularity, and marked transform.
4. Holomorphic reconstruction	`04_holomorphic_reconstruction.md`: pursue affine Bogomolny/twistor data or the induced elliptic-curve family; explain exactly which compatibility test for the same fixed field becomes simpler.
5. Alternative parent systems	`05_parent_systems.md`: first audit the restricted KYM and holomorphic-generator obstructions; then present a concrete auxiliary system only if its reduction enforces an otherwise missing physical condition.
6. Independent referee and synthesis	`06_referee.md` and `00_assessment.md`: test every claim against the fixed input and all global requirements; separate proof, formal expansion, numerical evidence, and conjecture.

Required reporting template

Each report should state:

1. The precise claim and its domain of validity.
2. Which quantities are fixed and which are unknown.
3. Physical assumptions versus optional ansatz restrictions.
4. A proof or explicit calculation, including reconstruction to the given Euclidean domain.
5. Boundary, period, nonvanishing, axis, circle-action, marking, and transform checks.
6. The strongest apparent objection, with its resolution or the remaining gap.
7. Primary references and a restrained account of what, if anything, is new.
8. Status: proved, formal, numerical, conjectural, or disproved under stated hypotheses.

Mandatory rejection tests

- Do not change the shape, metric, complex surface, or magnetic boundary to obtain an answer for this input.
- Do not call another HYM description of the same curvature a new physical solution.
- Do not replace strong QS by weak QS, a Killing symmetry, or one-angle dependence in arbitrary coordinates.
- Do not normalize u by a flux-dependent factor without checking (C), (Per), and the action.
- Do not infer full-volume torus foliation from one invariant torus, a KAM Cantor family, or a finite-order near-axis expansion.
- Do not suppress a nonzero current by changing the Hodge star.
- Do not infer exactness from a small residual; a certification theorem needs error and inverse bounds.
- Do not extrapolate a local, shell, cover, or abstract-flat result to this whole embedded solid torus.

9. Primary references and scope

The derivations in this brief are supplied for checking and carry no novelty claim. The references anchor terminology and identify relevant methods; none is cited as proving the full fixed-domain target.

- **[R1] Ted Jacobson**, *Guiding-centre Lagrangian and quasi-symmetry*, Journal of Plasma Physics 91 (2025), E46. DOI. Strong-QS formulation and its relation to other notions.
- **[R2] J. W. Burby, N. Kallinikos, R. S. MacKay**, *Characterization of admissible quasisymmetries*, Journal of Plasma Physics 91 (2025), E33. DOI. Restrictions on admissible generators; its hypotheses and additional vacuum constraints must be retained.
- **[R3] J. W. Burby, N. Kallinikos, R. S. MacKay**, *Some mathematics for quasi-symmetry*, Journal of Mathematical Physics 61 (2020), 093503. arXiv, author repository. Differential forms, symmetry, and global questions.
- **[R4] Matt Landreman, Elizabeth Paul**, *Magnetic Fields with Precise Quasisymmetry for Plasma Confinement*, Physical Review Letters 128 (2022), 035001. arXiv. Numerical approximation is distinct from a proof of exact strong QS throughout a volume.
- **[R5] Wrick Sengupta, Elizabeth J. Paul, Harold Weitzner, Amitava Bhattacharjee**, *Vacuum magnetic fields with exact quasisymmetry near a flux surface. Part 1. Solutions near an axisymmetric surface* (2021). arXiv. Check the expansion and geometry scope before using it for a global claim.
- **[R6a] Luis Álvarez-Cónsul, Mario Garcia-Fernandez, Oscar García-Prada**, *Gravitating vortices, cosmic strings, and the Kähler–Yang–Mills equations*, Communications in Mathematical Physics 351 (2017), 361–385. arXiv. The actual equivariant KYM-to-GV construction.
- **[R6b] Oscar García-Prada**, *Kähler–Yang–Mills Equations and Vortices*, SIGMA 20 (2024), 032. DOI, paper. Conventions for the coupled curvature equations.
- **[R7] Robin Roussel**, *Shape differentiation for Poincare maps of harmonic fields in toroidal domains.* arXiv:2406.08178. The one-dimensional space of normalized tangent harmonic fields and field-line return maps are relevant here; shape variation is outside this assignment.
- **[R8] J. W. Burby, N. Kallinikos, R. S. MacKay**, *Generalized Grad–Shafranov equation for non-axisymmetric MHD equilibria*, Physics of Plasmas 27 (2020), 102504. DOI, author repository. A reduction framework whose averaged equations and nonzero-pressure-gradient assumptions must not be silently transferred to pointwise vacuum.

- **[R9] Nigel Hitchin**, *Monopoles and geodesics*, Communications in Mathematical Physics 83 (1982), 579–602. Author's publications with paper link. The minitwistor viewpoint; this toroidal boundary problem requires separate global analysis.
- **[R10] R. S. MacKay**, *Differential forms for plasma physics*, Journal of Plasma Physics (2020). DOI. Background on magnetic forms and local versus global potentials.

Before making an openness, nonexistence, or novelty claim, check current primary literature. This brief does not assert a universal nonexistence theorem for exact nonaxisymmetric vacuum QS. Its proved negative statements are the specified ansatz obstructions and the explicit local test.

10. The question to return an answer to

For the **given** Ω, compute or characterize its unique normalized field B_P. Does it admit a solution of (FQS) whose reconstructed generator and moment satisfy every global condition in Section 2?

Equivalently, does its fixed holomorphic HYM curvature admit the additional real symmetry data in (HQS) with those global conditions?

A successful reduction must answer that compatibility question, or prove a meaningful necessary or sufficient condition for it. A solution on a different domain or complex surface answers a different question.